

Transverse and longitudinal waves

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

	QUESTION	ANSWER	✓
1	State what waves transfer.	Energy	☐
2	State what waves do not transfer.	Matter	☐
3	Name the material that a wave travels through.	A medium	☐
4	Name a repeated back-and-forth motion.	An oscillation	☐
5	State what oscillates in a water wave.	The water particles (up and down)	☐
6	State what oscillates in a sound wave in air.	The air particles	☐
7	State what oscillates in a light wave.	Electric and magnetic fields	☐
8	Name the two types of wave.	Transverse waves and longitudinal waves	☐
9	What is a transverse wave?	A wave in which the oscillations are perpendicular to the direction of energy transfer.	☐
10	What is a longitudinal wave?	A wave in which the oscillations are parallel to the direction of energy transfer.	☐
11	What type of wave is sound?	Longitudinal	☐
12	What type of wave is light?	Transverse	☐
13	State what labels A and B show on this longitudinal wave. 	A: a compression B: a rarefaction	☐

QUESTION		ANSWER	✓
14	Give an example of a transverse wave.	Ripples on water (or light)	☐
15	Give an example of a longitudinal wave.	Sound waves in air	☐
16	Describe evidence that water waves transfer energy without moving the water along.	A floating object bobs up and down but is not carried along.	☐